

Data Mining

Lab - 1

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Introduction to Pandas Library Function:

Step-1 Import the pandas Libraries

```
In [1]: import pandas as pd
```

Step-2 Import the dataset from this:....

```
In [ ]:
```

Step-3 Read csv or excel File

```
In [2]: df = pd.read_csv("../data/titanic.csv")
```

Step-4 Print Data from csv or excel File

```
In [3]: df
```

Out[3]:

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500	NaN	S
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2833	C85	C
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250	NaN	S
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.1000	C123	S
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450	8.0500	NaN	S
...	...	...	...	...	...	...	...	...	...	...	...	...
886	887	0	2	Montvila, Rev. Juozas	male	27.0	0	0	211536	13.0000	NaN	S
887	888	1	1	Graham, Miss. Margaret Edith	female	19.0	0	0	112053	30.0000	B42	S
888	889	0	3	Johnston, Miss. Catherine Helen "Carrie"	female	NaN	1	2	W./C. 6607	23.4500	NaN	S
889	890	1	1	Behr, Mr. Karl Howell	male	26.0	0	0	111369	30.0000	C148	C
890	891	0	3	Dooley, Mr. Patrick	male	32.0	0	0	370376	7.7500	NaN	Q

891 rows × 12 columns

Step-5 See the First 10 Rows

In [4]:

df.head(10)

Out[4]:

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500	NaN	S
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2833	C85	C
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250	NaN	S
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.1000	C123	S
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450	8.0500	NaN	S
5	6	0	3	Moran, Mr. James	male	NaN	0	0	330877	8.4583	NaN	Q
6	7	0	1	McCarthy, Mr. Timothy J	male	54.0	0	0	17463	51.8625	E46	S
7	8	0	3	Palsson, Master. Gosta Leonard	male	2.0	3	1	349909	21.0750	NaN	S
8	9	1	3	Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)	female	27.0	0	2	347742	11.1333	NaN	S
9	10	1	2	Nasser, Mrs. Nicholas (Adele Achem)	female	14.0	1	0	237736	30.0708	NaN	C

Step-6 See the Last 10 Rows

In [5]:

df.tail(10)

Out[5]:

	PassengerId	Survived	Pclass		Name	Sex	Age	SibSp	Parch		Ticket	Fare	Cabin	Embarked	
	881	882	0	3		Markun, Mr. Johann	male	33.0	0	0		349257	7.8958	NaN	S
	882	883	0	3		Dahlberg, Miss. Gerda Ulrika	female	22.0	0	0		7552	10.5167	NaN	S
	883	884	0	2		Banfield, Mr. Frederick James	male	28.0	0	0	C.A./SOTON	34068	10.5000	NaN	S
	884	885	0	3		Sutehall, Mr. Henry Jr	male	25.0	0	0	SOTON/OQ	392076	7.0500	NaN	S
	885	886	0	3		Rice, Mrs. William (Margaret Norton)	female	39.0	0	5		382652	29.1250	NaN	Q
	886	887	0	2		Montvila, Rev. Juozas	male	27.0	0	0		211536	13.0000	NaN	S
	887	888	1	1		Graham, Miss. Margaret Edith	female	19.0	0	0		112053	30.0000	B42	S
	888	889	0	3		Johnston, Miss. Catherine Helen "Carrie"	female	NaN	1	2	W./C.	6607	23.4500	NaN	S
	889	890	1	1		Behr, Mr. Karl Howell	male	26.0	0	0		111369	30.0000	C148	C
	890	891	0	3		Dooley, Mr. Patrick	male	32.0	0	0		370376	7.7500	NaN	Q

Step-7 Data type of each columns

In [6]: df.dtypes

Out[6]: PassengerId int64
Survived int64
Pclass int64
Name object
Sex object
Age float64
SibSp int64
Parch int64
Ticket object
Fare float64
Cabin object
Embarked object
dtype: object

Step-8 Display Summary Information

In [7]: df.info()

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
#   Column      Non-Null Count  Dtype
---  -
0   PassengerId  891 non-null    int64
1   Survived     891 non-null    int64
2   Pclass       891 non-null    int64
3   Name         891 non-null    object
4   Sex          891 non-null    object
5   Age          714 non-null    float64
6   SibSp        891 non-null    int64
7   Parch        891 non-null    int64
8   Ticket       891 non-null    object
9   Fare         891 non-null    float64
10  Cabin        204 non-null    object
11  Embarked     889 non-null    object
dtypes: float64(2), int64(5), object(5)
memory usage: 83.7+ KB
```

Step-9 Access a specific column

```
In [8]: df["Name"]

Out[8]: 0      Braund, Mr. Owen Harris
1  Cumings, Mrs. John Bradley (Florence Briggs Th...
2      Heikkinen, Miss. Laina
3  Futrelle, Mrs. Jacques Heath (Lily May Peel)
4      Allen, Mr. William Henry
...
886      Montvila, Rev. Juozas
887      Graham, Miss. Margaret Edith
888  Johnston, Miss. Catherine Helen "Carrie"
889      Behr, Mr. Karl Howell
890      Dooley, Mr. Patrick
Name: Name, Length: 891, dtype: object
```

Step-10 Access rows by their integer location

```
In [9]: df.iloc[7]

Out[9]: PassengerId      8
Survived      0
Pclass      3
Name      Palsson, Master. Gosta Leonard
Sex      male
Age      2.0
SibSp      3
Parch      1
Ticket      349909
Fare      21.075
Cabin      NaN
Embarked      S
Name: 7, dtype: object
```

Step-11 Delete a specific Column

```
In [10]: # df=df.drop(columns=['Cabin'])
df.drop("Cabin", axis=1,inplace=True)
```

```
df
```

Out[10]:

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Embarked	
	0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500	S
	1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2833	C
	2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250	S
	3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.1000	S
	4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450	8.0500	S
	...	...	...	...	...	...	...	...	...	...	...	...
	886	887	0	2	Montvila, Rev. Juozas	male	27.0	0	0	211536	13.0000	S
	887	888	1	1	Graham, Miss. Margaret Edith	female	19.0	0	0	112053	30.0000	S
	888	889	0	3	Johnston, Miss. Catherine Helen "Carrie"	female	NaN	1	2	W./C. 6607	23.4500	S
	889	890	1	1	Behr, Mr. Karl Howell	male	26.0	0	0	111369	30.0000	C
	890	891	0	3	Dooley, Mr. Patrick	male	32.0	0	0	370376	7.7500	Q

891 rows × 11 columns

Step-12 Create a new Column

```
In [11]: df["isName"] = ~df["Name"].isnull()  
df
```

Out[11]:

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Embarked	isName	
	0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500	S	True
	1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2833	C	True
	2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250	S	True
	3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.1000	S	True
	4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450	8.0500	S	True
	...	...	...	...	...	...	...	...	...	...	...	...	...
	886	887	0	2	Montvila, Rev. Juozas	male	27.0	0	0	211536	13.0000	S	True
	887	888	1	1	Graham, Miss. Margaret Edith	female	19.0	0	0	112053	30.0000	S	True
	888	889	0	3	Johnston, Miss. Catherine Helen "Carrie"	female	NaN	1	2	W./C. 6607	23.4500	S	True
	889	890	1	1	Behr, Mr. Karl Howell	male	26.0	0	0	111369	30.0000	C	True
	890	891	0	3	Dooley, Mr. Patrick	male	32.0	0	0	370376	7.7500	Q	True

891 rows × 12 columns

Step-13 Perform Condition Selection on DataFrame

```
In [12]: df[(df["Pclass"] > 1) & (df["isName"] == True)]
```

Out[12]:

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Embarked	isName	
	0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500	S	True
	2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250	S	True
	4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450	8.0500	S	True
	5	6	0	3	Moran, Mr. James	male	NaN	0	0	330877	8.4583	Q	True
	7	8	0	3	Palsson, Master. Gosta Leonard	male	2.0	3	1	349909	21.0750	S	True
	...	...	...	...	...	...	...	...	...	...	...	...	...
	884	885	0	3	Sutehall, Mr. Henry Jr	male	25.0	0	0	SOTON/OQ 392076	7.0500	S	True
	885	886	0	3	Rice, Mrs. William (Margaret Norton)	female	39.0	0	5	382652	29.1250	Q	True
	886	887	0	2	Montvila, Rev. Juozas	male	27.0	0	0	211536	13.0000	S	True
	888	889	0	3	Johnston, Miss. Catherine Helen "Carrie"	female	NaN	1	2	W./C. 6607	23.4500	S	True
	890	891	0	3	Dooley, Mr. Patrick	male	32.0	0	0	370376	7.7500	Q	True

675 rows × 12 columns

Step-14 Compute the sum of value

In [13]: df["Fare"].sum()

Out[13]: 28693.9493

Step-15 Compute the mean of value

In [14]: df["Fare"].mean()

Out[14]: 32.204207968574636

Step-16 Count non-null value (column)

In []: df.count()

Out[]: PassengerId 891
Survived 891
Pclass 891
Name 891
Sex 891
Age 714
SibSp 891
Parch 891
Ticket 891
Fare 891
Embarked 889
isName 891
dtype: int64

Step-17 Find Minimun or Maximum values

In []: `print(df["Fare"].min())`

`print(df["Fare"].max())`

0.0

512.3292